```
import pandas as pd
import matplotlib.pyplot as plt

# -------------------------------
# SETTINGS
# -------------------------------
SHEET_URL =
"https://docs.google.com/spreadsheets/d/e/2PACX-1vSdpnoLmX4cyxQAm799-uRYHTw0otbF1
KPBqlu6398noWbDb_d3hYbv22PBJHW0zmahdszeJzVXNp6C/pub?output=csv"

REGISTERED = 250
COST_PER_PLATE = 60
CO2_PER_PLATE = 2.5
WATER_PER_PLATE = 18

# -------------------------------
# LOAD DATA
# -------------------------------
def load_data():
    try:
        df = pd.read_csv(SHEET_URL)
    except Exception as e:
        print("❌ Error loading data:", e)
        return None, None, None, None, None

    # Clean column names
    df.columns = df.columns.str.lower().str.strip()

    def find_col(keyword):
        for c in df.columns:
            if keyword in c:
                return c
        return None

    id_col = find_col("id")
    b_col  = find_col("breakfast")
    l_col  = find_col("lunch")
    d_col  = find_col("dinner")

    # ⚠️ Important fix: handle missing columns
    if not all([id_col, b_col, l_col, d_col]):
        print("❌ Required columns not found in sheet!")
        print("Found columns:", df.columns.tolist())
```

```
        return None, None, None, None, None

    # Remove duplicates (keep latest response)
    if "timestamp" in df.columns:
        df = df.sort_values("timestamp").drop_duplicates(subset=id_col, keep="last")

    return df, id_col, b_col, l_col, d_col


# -------------------------------
# VIEW RECORDS
# -------------------------------
def view_records(df, id_col, b_col, l_col, d_col):
    if df is None:
        return

    print("\n📋 Attendance Records:")
    print(f"{'ID':10} {'Breakfast':10} {'Lunch':10} {'Dinner':10}")
    print("-" * 50)

    for _, row in df.iterrows():
        print(f"{str(row[id_col])[:10]:10} "
              f"{str(row[b_col])[:10]:10} "
              f"{str(row[l_col])[:10]:10} "
              f"{str(row[d_col])[:10]:10}")


# -------------------------------
# COUNT
# -------------------------------
def count_meals(df, b_col, l_col, d_col):
    if df is None:
        return 0, 0, 0

    breakfast = (df[b_col].astype(str).str.lower() == "yes").sum()
    lunch     = (df[l_col].astype(str).str.lower() == "yes").sum()
    dinner    = (df[d_col].astype(str).str.lower() == "yes").sum()

    return breakfast, lunch, dinner


# -------------------------------
# SAVINGS
# -------------------------------
```

```
def show_savings(b, l, d):
    saved_b = REGISTERED - b
    saved_l = REGISTERED - l
    saved_d = REGISTERED - d

    total_saved = saved_b + saved_l + saved_d

    money = total_saved * COST_PER_PLATE
    co2   = total_saved * CO2_PER_PLATE
    water = total_saved * WATER_PER_PLATE

    print("\n💰 SAVINGS REPORT")
    print(f"Breakfast saved: {saved_b}")
    print(f"Lunch saved: {saved_l}")
    print(f"Dinner saved: {saved_d}")
    print(f"Total plates saved: {total_saved}")
    print(f"Money saved: ₹ {money}")
    print(f"CO₂ saved: {co2} kg")
    print(f"Water saved: {water} liters")


# -------------------------------
# GRAPH
# -------------------------------
def show_graph(b, l, d):
    meals = ["Breakfast", "Lunch", "Dinner"]
    values = [b, l, d]

    plt.figure()
    bars = plt.bar(meals, values)

    plt.title("Meal Attendance")
    plt.xlabel("Meals")
    plt.ylabel("Students")

    for bar in bars:
        plt.text(
            bar.get_x() + bar.get_width() / 2,
            bar.get_height(),
            str(int(bar.get_height())),
            ha='center',
            va='bottom'
        )
```

```
    plt.show()


# -------------------------------
# MENU
# -------------------------------
def menu():
    while True:
        print("\n====== SMART MESS SYSTEM ======")
        print("1. View Records")
        print("2. Count Meals")
        print("3. Show Savings")
        print("4. Show Graph")
        print("5. Exit")

        choice = input("Enter choice: ")

        df, id_col, b_col, l_col, d_col = load_data()

        if df is None:
            continue

        b, l, d = count_meals(df, b_col, l_col, d_col)

        if choice == "1":
            view_records(df, id_col, b_col, l_col, d_col)

        elif choice == "2":
            print("\n📊 COUNT")
            print("Breakfast:", b)
            print("Lunch:", l)
            print("Dinner:", d)

        elif choice == "3":
            show_savings(b, l, d)

        elif choice == "4":
            show_graph(b, l, d)

        elif choice == "5":
            print("👋 Exiting...")
            break

        else:
```

```
        print("❌ Invalid choice")


# -------------------------------
# RUN
# -------------------------------
menu()
```

```
import pandas as pd
import matplotlib.pyplot as plt

# -------------------------------
# SETTINGS
# -------------------------------
SHEET_URL =
"https://docs.google.com/spreadsheets/d/e/2PACX-1vSdpnoLmX4cyxQAm799-uRYHTw0otbF1
KPBqlu6398noWbDb_d3hYbv22PBJHW0zmahdszeJzVXNp6C/pub?output=csv"

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def load_data():
    try:
        df = pd.read_csv(SHEET_URL)
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        print("❌ Error loading data:", e)
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    # Clean column names
    df.columns = df.columns.str.lower().str.strip()

    def find_col(keyword):
        for c in df.columns:
            if keyword in c:
```

```
            return c
        return None

    id_col = find_col("id")
    b_col  = find_col("breakfast")
    l_col  = find_col("lunch")
    d_col  = find_col("dinner")

    # ⚠️ Important fix: handle missing columns
    if not all([id_col, b_col, l_col, d_col]):
        print("❌ Required columns not found in sheet!")
        print("Found columns:", df.columns.tolist())
        return None, None, None, None, None

    # Remove duplicates (keep latest response)
    if "timestamp" in df.columns:
        df = df.sort_values("timestamp").drop_duplicates(subset=id_col, keep="last")

    return df, id_col, b_col, l_col, d_col


# -------------------------------
# VIEW RECORDS
# -------------------------------
def view_records(df, id_col, b_col, l_col, d_col):
    if df is None:
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    print("\n📋 Attendance Records:")
    print(f"{'ID':10} {'Breakfast':10} {'Lunch':10} {'Dinner':10}")
    print("-" * 50)

    for _, row in df.iterrows():
        print(f"{str(row[id_col])[:10]:10} "
              f"{str(row[b_col])[:10]:10} "
              f"{str(row[l_col])[:10]:10} "
              f"{str(row[d_col])[:10]:10}")


# -------------------------------
# COUNT
# -------------------------------
def count_meals(df, b_col, l_col, d_col):
    if df is None:
```

```
        return 0, 0, 0

    breakfast = (df[b_col].astype(str).str.lower() == "yes").sum()
    lunch     = (df[l_col].astype(str).str.lower() == "yes").sum()
    dinner    = (df[d_col].astype(str).str.lower() == "yes").sum()

    return breakfast, lunch, dinner


# -------------------------------
# SAVINGS
# -------------------------------
def show_savings(b, l, d):
    saved_b = REGISTERED - b
    saved_l = REGISTERED - l
    saved_d = REGISTERED - d

    total_saved = saved_b + saved_l + saved_d

    money = total_saved * COST_PER_PLATE
    co2   = total_saved * CO2_PER_PLATE
    water = total_saved * WATER_PER_PLATE

    print("\n💰 SAVINGS REPORT")
    print(f"Breakfast saved: {saved_b}")
    print(f"Lunch saved: {saved_l}")
    print(f"Dinner saved: {saved_d}")
    print(f"Total plates saved: {total_saved}")
    print(f"Money saved: ₹ {money}")
    print(f"CO₂ saved: {co2} kg")
    print(f"Water saved: {water} liters")


# -------------------------------
# GRAPH
# -------------------------------
def show_graph(b, l, d):
    meals = ["Breakfast", "Lunch", "Dinner"]
    values = [b, l, d]

    plt.figure()
    bars = plt.bar(meals, values)

    plt.title("Meal Attendance")
```

```
    plt.xlabel("Meals")
    plt.ylabel("Students")

    for bar in bars:
        plt.text(
            bar.get_x() + bar.get_width() / 2,
            bar.get_height(),
            str(int(bar.get_height())),
            ha='center',
            va='bottom'
        )

    plt.show()


# -------------------------------
# MENU
# -------------------------------
def menu():
    while True:
        print("\n====== SMART MESS SYSTEM ======")
        print("1. View Records")
        print("2. Count Meals")
        print("3. Show Savings")
        print("4. Show Graph")
        print("5. Exit")

        choice = input("Enter choice: ")

        df, id_col, b_col, l_col, d_col = load_data()

        if df is None:
            continue

        b, l, d = count_meals(df, b_col, l_col, d_col)

        if choice == "1":
            view_records(df, id_col, b_col, l_col, d_col)

        elif choice == "2":
            print("\n📊 COUNT")
            print("Breakfast:", b)
            print("Lunch:", l)
            print("Dinner:", d)
```

```
    elif choice == "3":
        show_savings(b, l, d)

    elif choice == "4":
        show_graph(b, l, d)

    elif choice == "5":
        print("👋 Exiting...")
        break

    else:
        print("❌ Invalid choice")


# ------------------------------
# RUN
# ------------------------------
menu()
```

```
====== SMART MESS SYSTEM ======
1. View Records
2. Count Meals
3. Show Savings
4. Show Graph
5. Exit

Enter choice:  1

📋 Attendance Records:
ID          Breakfast  Lunch      Dinner
--------------------------------------------------
573         Yes        Yes        Yes
233         Yes        Yes        Yes
235         Yes        Yes        Yes
225         No         Yes        Yes
226         No         Yes        Yes
227         Yes        Yes        Yes
228         No         Yes        Yes
229         No         Yes        Yes
230         No         Yes        Yes
231         Yes        Yes        Yes
```

232	Yes	Yes	Yes
236	Yes	Yes	Yes
237	Yes	Yes	Yes
238	No	Yes	Yes
239	No	Yes	Yes
240	Yes	NO	Yes
345	Yes	Yes	No
234	Yes	NO	Yes
879	No	Yes	Yes
765	Yes	Yes	Yes
241	No	Yes	Yes
242	No	Yes	Yes
243	Yes	Yes	Yes
244	No	Yes	Yes
400	Yes	No	Yes
401	No	No	Yes
402	Yes	No	No
500	Yes	No	Yes
403	Yes	No	No
245	No	Yes	Yes
501	No	Yes	Yes
246	No	Yes	Yes
502	No	Yes	Yes
404	Yes	Yes	Yes
503	Yes	Yes	Yes
247	No	Yes	Yes
300	No	Yes	No
405	No	Yes	Yes
248	Yes	Yes	Yes
504	Yes	Yes	No
505	Yes	Yes	No
506	Yes	Yes	Yes
249	No	Yes	Yes
507	Yes	Yes	Yes
406	Yes	Yes	No
250	No	Yes	Yes
508	Yes	Yes	Yes
407	Yes	No	No
201	No	Yes	Yes
509	Yes	Yes	Yes
408	No	No	Yes
202	Yes	Yes	Yes
409	Yes	Yes	Yes
510	Yes	No	Yes
203	Yes	Yes	No
410	No	Yes	No
204	Yes	Yes	Yes
411	No	Yes	Yes
511	Yes	Yes	Yes

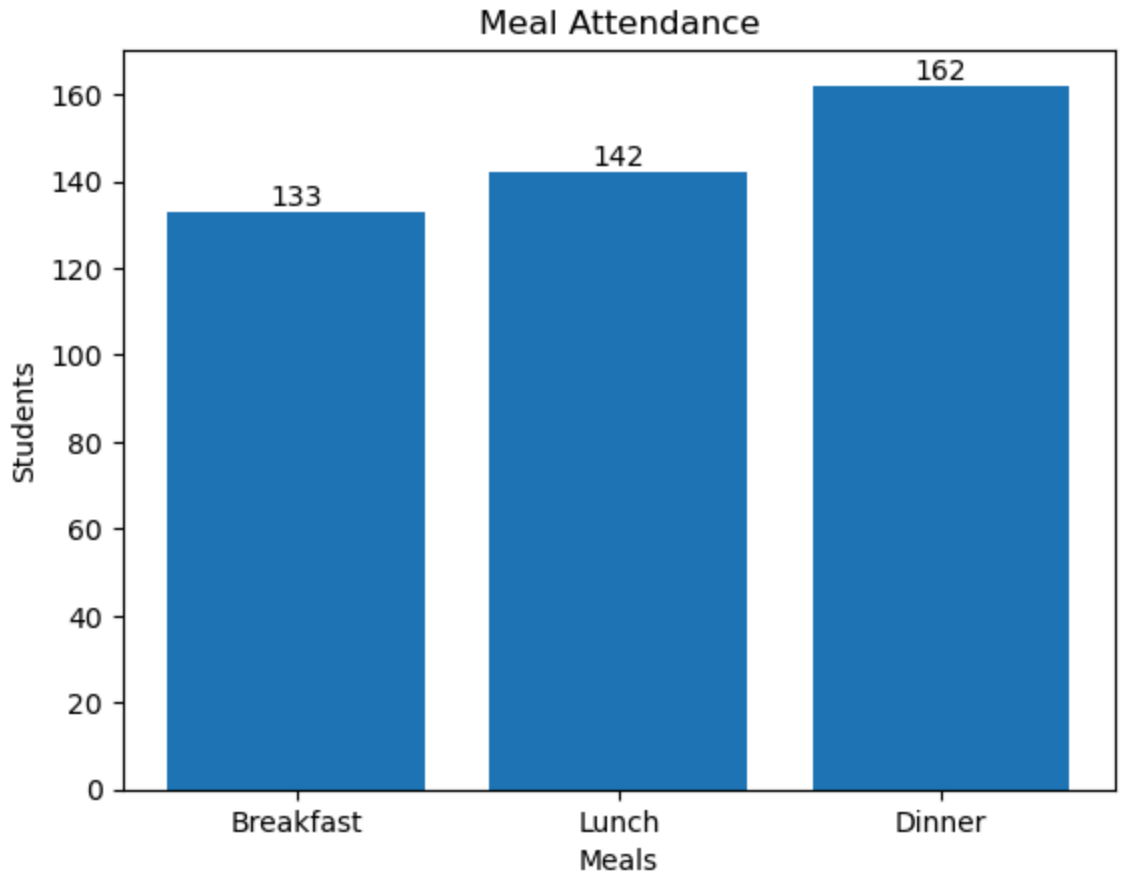
205	No	Yes	Yes
412	Yes	Yes	No
512	Yes	No	Yes
413	Yes	Yes	No
513	Yes	No	Yes
206	Yes	Yes	Yes
414	No	Yes	Yes
514	Yes	No	Yes
515	No	No	Yes
516	Yes	No	Yes
207	No	No	Yes
517	Yes	No	Yes
208	Yes	Yes	Yes
209	No	Yes	Yes
518	No	Yes	Yes
519	No	Yes	Yes
210	Yes	Yes	Yes
520	Yes	Yes	Yes
211	No	Yes	Yes
521	Yes	Yes	Yes
212	Yes	Yes	Yes
522	No	Yes	Yes
213	Yes	Yes	Yes
301	Yes	No	Yes
214	No	Yes	Yes
415	Yes	Yes	Yes
215	No	Yes	Yes
416	Yes	Yes	No
302	No	No	Yes
216	No	Yes	Yes
417	No	Yes	Yes
418	No	No	Yes
217	No	Yes	Yes
523	Yes	Yes	Yes
419	No	No	Yes
524	Yes	Yes	Yes
218	No	Yes	Yes
420	No	No	Yes
525	Yes	Yes	Yes
219	No	Yes	Yes
421	No	No	Yes
526	Yes	No	Yes
422	No	Yes	Yes
303	Yes	Yes	Yes
220	Yes	Yes	Yes
527	Yes	Yes	Yes
423	No	Yes	Yes
528	No	Yes	Yes
424	No	Yes	Yes

304	No	No	No
529	Yes	Yes	Yes
221	No	Yes	Yes
425	No	No	No
305	Yes	No	Yes
426	Yes	Yes	No
222	Yes	Yes	Yes
306	Yes	Yes	No
223	Yes	Yes	Yes
428	Yes	Yes	No
530	Yes	No	Yes
224	Yes	Yes	Yes
429	No	No	Yes
307	Yes	Yes	No
531	Yes	Yes	Yes
430	Yes	Yes	No
532	Yes	Yes	Yes
431	Yes	No	Yes
308	No	No	Yes
533	Yes	No	Yes
432	Yes	No	Yes
433	No	No	No
309	Yes	Yes	Yes
535	Yes	Yes	Yes
435	Yes	No	Yes
310	Yes	No	Yes
536	Yes	No	Yes
436	No	No	Yes
537	Yes	Yes	Yes
311	Yes	No	Yes
437	Yes	No	Yes
538	Yes	Yes	Yes
438	Yes	Yes	No
539	Yes	Yes	Yes
312	Yes	Yes	Yes
439	No	Yes	No
540	Yes	No	Yes
440	No	No	No
313	No	No	No
541	Yes	Yes	Yes
441	Yes	Yes	Yes
542	Yes	Yes	Yes
442	Yes	Yes	Yes
314	Yes	Yes	No
543	No	Yes	Yes
443	Yes	Yes	Yes
544	Yes	Yes	Yes
315	Yes	Yes	Yes
444	Yes	No	Yes

545	Yes	Yes	Yes
316	Yes	Yes	Yes
445	Yes	Yes	Yes
546	Yes	No	Yes
446	Yes	Yes	Yes
547	No	Yes	Yes
317	No	Yes	No
447	No	Yes	Yes
448	Yes	Yes	Yes
318	Yes	Yes	Yes
548	Yes	No	Yes
449	Yes	Yes	Yes
319	No	Yes	No
549	Yes	Yes	Yes
450	No	No	Yes
550	Yes	Yes	Yes
320	Yes	No	Yes
321	Yes	Yes	Yes
324	No	No	No
322	Yes	No	Yes
323	Yes	No	Yes
325	Yes	No	Yes
326	Yes	Yes	No
327	Yes	Yes	No
328	Yes	No	Yes
329	Yes	Yes	No
330	Yes	No	No
331	Yes	Yes	No
332	Yes	No	No
333	Yes	Yes	Yes
334	Yes	No	No
335	Yes	Yes	No
336	Yes	Yes	No
337	Yes	Yes	No
338	No	Yes	Yes
339	Yes	No	Yes
340	Yes	Yes	Yes
341	No	No	Yes
342	Yes	Yes	Yes
343	No	No	No
344	No	Yes	No
346	Yes	No	Yes
347	Yes	Yes	Yes
348	Yes	Yes	Yes
349	Yes	No	No
350	Yes	No	Yes
123	Yes	Yes	Yes

====== SMART MESS SYSTEM ======

```
1. View Records
2. Count Meals
3. Show Savings
4. Show Graph
5. Exit

Enter choice:  2

📊 COUNT
Breakfast: 133
Lunch: 142
Dinner: 162

====== SMART MESS SYSTEM ======
1. View Records
2. Count Meals
3. Show Savings
4. Show Graph
5. Exit

Enter choice:  3

💰 SAVINGS REPORT
Breakfast saved: 117
Lunch saved: 108
Dinner saved: 88
Total plates saved: 313
Money saved: ₹ 18780
CO₂ saved: 782.5 kg
Water saved: 5634 liters

====== SMART MESS SYSTEM ======
1. View Records
2. Count Meals
3. Show Savings
4. Show Graph
5. Exit

Enter choice:  4
```




```
====== SMART MESS SYSTEM ======
1. View Records
2. Count Meals
3. Show Savings
4. Show Graph
5. Exit
```